

SIMATS
ENGINEERING

Secure Exit

SIMATS Online Programming
Lab

JAVA PROGRAMMING

PENUGONDA
JEEVAN KUMAR
192472082RD

CO3-AT3-Assignment

View Only

[← Back](#)

QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Hospital Emergency System –
Thread Priority
10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order

Q1 Banking Transaction System – Exception Handling

10 marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario Expected Result Purpose

10000 3000 Normal case

10000 12000 Boundary / exception case

10000 -500 Invalid input

abc 3000 Invalid input

10 marks

5/5 answered

Your Code

```
1 class Main{
2     public static void main(String[] args){
3         double balance=10000,amount=3000;
4         try{
5             if(amount<=0)throw new IllegalArgumentException("Invalidamou
6             if(amount > balance)throw new InsufficientBalanceException("
7             System.out.println("Withdrawal successful");
8         }catch(Exception e){
9             System.out.println(e.getMessage());
10        }finally{
11            System.out.println("Transaction completed");
12        }
13    }
14 }
15 class InsufficientBalanceException extends Exception{
16     InsufficientBalanceException(String s){
17         super(s);
18     }
19 }
```

Input

stdin input

Output

```
Withdrawal successful  
Transaction completed
```

✓ Submitted

© 2026 **SIMATS Online Lab**. All rights reserved.
